

Terra Preta: Fermentation to Soil

Carpathian OER Commons — Land, Forest, Food Module

Michel Garand | 2026-09-29 | Version v0.2 | CC BY-SA 4.0

Maturity: L1 — designed and costed, untested. Every cost line is an estimate or a listed price, labelled as such in Section 6; three are still costed at the working paper's 25 people. Nothing has been sealed, bedded, or matured. Read every figure below at this level and no higher.

At this site, nothing in this module is bought or built before the district sanitary service has answered in writing whether matured material from this stream may go on the site's own trees (Section 10). Until then the module is a design, and says so.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Converted from the sanitation working paper, EN v0.2, Sections 3 (the bucket exchange), 4, 5, 7, 8.3, 8.4, 10, and 11. The toilets — seat, bucket, cubicle, vent, and the char bin at the seat — stay with the sanitation toilets row; the greywater, reed bed, and pond with theirs. The paper is sized at 25 people; every quantity here that depends on group size is recalculated at 34, and the paper's own omissions are corrected in Section 3.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

v0.2, 2026-09-29: a note on permits and local rules added at the head of the module; nothing else changed.

1. Purpose

The toilet stream made into soil for trees: everything that leaves the bucket toilets, and the kitchen's cooked food, plate waste, and grease-trap solids, sealed and fermented, worked by worms, and matured a year, then used on trees, hedgerows, windbreaks, and nursery stock only. Simple where it is handled, careful where it matters.

Six rules. One stream: nothing is separated; urine goes in with the rest, and the kitchen's waste goes into the same vessels. Carbon holds it: char and a carbon cover at every use, because char alone cannot hold the urine. Ferment first, worms second, time last: sealed lactic fermentation for at least six weeks, worm beds fed layered and bulked, then a year in a closed bay. The clock counts from the last addition: each bay is filled as a batch and closed, and its year runs from the day it closes. Trees only: never on

vegetable beds, never on anything harvested raw within the season, and nothing in the month before a fruit harvest. One responsible adult, paid: buckets, vessels, and worm beds are handled by one named adult and a deputy, not a rota; the work is a cost line, and toloka is not its name.

Connections. Depends on: the sanitation toilets row, which delivers the full buckets and sets the char and cover at the seat; the biochar module, for char; the compost module, which hands over cooked food, plate waste, and grease-trap solids; the kitchen and dining module for those volumes; the written answer of the district sanitary service; the Phase 1 walk, with every distance in Section 4 measured with a tape. Feeds: the trees, hedgerows, windbreaks, and the tree nursery module's stock; the planting module. Rinse water from the buckets goes to the septic tank, in the greywater module, never to the greywater line.

Patterns. Grows from: not yet named. What this valley did with a household's own waste, and whether anything of it was returned to the land, are questions for the elders; the literature found nothing. Feeds: not yet named. Resonates with: 178 Compost — toilets and kitchen waste made into soil together, as Alexander asks, for the trees; 170 Fruit Trees — trees, hedgerows, and nursery stock as the ground the soil goes to; 104 Site Repair — store, beds, and bays on the poorer ground; 25 Access to Water — the protective strip keeps them out of the belt along the stream and the pond; 64 Pools and Streams — a sealed base and covers keep a wet stream off running water; 127 Intimacy Gradient — the store as the most withdrawn place on the site. In tension with: 18 Network of Learning — the work is shown only as far as the worm beds, and never the vessels; 120 Paths and Goals — the daily route of the buckets is a working path, kept short and apart from the day's movement.

2. Context of validity

Condition	Range this design holds for
Climate	[set] — mountain; worm beds banked in winter
Altitude	[set]
Group size	capacity: 34 people, the example below
Days of use	use_days_per_year: 120, the example below
Session	session_length: 20 days
Toilets	bucket toilets carrying two-thirds of uses
Store	above freezing all year; shaded; lockable
Ground	level, firm, outside every distance

Condition	Range this design holds for
Receiving ground	trees, hedgerows, nursery stock, [set]

If the sanitary service requires plumbed units for the children's side (sanitation paper, Section 9), bucket_share falls to the share of uses that staff, guests, and the outdoor working day make, [set]; every bucket figure below falls with it, and the kitchen's share does not.

3. Sizing method

What you must measure yourself before using this module:

- **Your bucket volume** — the full buckets of a session, measured by the litre as they are tipped, against the head count at capacity.
- **Your kitchen's waste** — cooked food, plate waste, and grease-trap solids, weighed for a full session.
- **Your cover** — how much cover a use takes to leave no smell, found in the first week, as Jenkins advises.
- **Your pH** — each vessel's pH at sealing and at opening, to set the starter and the carbohydrate.
- **Your receiving ground** — the trees, hedgerows, windbreaks, and nursery beds that can take a year's soil, counted on the walk.
- **Your distances** — as in Section 4, measured with a tape.

Our example: capacity, 34 people; bucket_share, two-thirds; urine_per_person, 1.4 L a day; faeces_per_person, 0.128 kg a day; char_at_seat, 0.5 L a day; cover_at_seat, 0.8 L a day, [assumed]; meals_per_day, 3, [assumed]; food_waste_per_portion, 58 to 129.5 g; use_days_per_year, 120; compost_volume_ratio, 0.33 to 0.75. Values from the site parameters file, held in the workshop.

3.1 What goes into the vessels, a day at capacity

Stream	Calculation	Our example
People at the buckets	$34 \times \text{two-thirds}$	about 22.7
Urine	$22.7 \times 1.4 \text{ L}$	about 31.7 L
Faeces	$22.7 \times 0.128 \text{ kg}$	about 2.9 L
Char at the seat	$22.7 \times 0.5 \text{ L}$	about 11.3 L
Cover at the seat	$22.7 \times 0.8 \text{ L}$	about 18.1 L

Stream	Calculation	Our example
Buckets, a day	sum	about 64 L
Cooked food and plates	$102 \times 58\text{--}129.5 \text{ g}$	5.9–13.2 kg
Same, by volume	$\div 737 \text{ kg per m}^3$	about 8–18 L
Into the vessels, a day	sum	about 72–85 L

The upper figure takes the paper's faeces figure, 0.25 L a person, instead of the median; the difference is under 3 L a day.

Where the paper's figures change. The paper counted faeces and char only, about 19 L a day at 25 people. It left out the urine, which it keeps in the stream (Section 3 of the paper) and which is the largest part of it; it did not apply its own share of two-thirds of uses to the buckets; and its faeces figure, 0.25 L a person a day, is about twice the median of 128 g wet a person a day, at 75 percent water, in Rose and colleagues' review (2015), which also gives 1.4 L of urine a person a day as the median. Faeces are taken here at about 1 kg a litre, [assumed].

The cover. Char holds about 1 to 4 mg of nitrogen a gram (Bischak and colleagues, 2024) or less (Fidel and colleagues, 2018), and urine carries a median 11 g of nitrogen a person a day (Rose and colleagues, 2015): no scoop of char can hold it. Joseph Jenkins's humanure practice takes urine and faeces together under a carbon cover, and finds that one 20 L receptacle serves one adult for a week with the right cover (Humanure Sanitation, 2009) — about 2.9 L a person a day in all. cover_at_seat, 0.8 L, is what is left of that after urine, faeces, and char, and is [assumed] until the first week sets it. Sawdust, dry leaves from the yard, or shredded straw.

The kitchen. Food service in Finland wasted 58 to 129.5 g a portion across sectors, kitchen, serving, and plate together (Silvennoinen and colleagues, Waste Management, 2019). Portions are capacity $\times$ meals_per_day. Kitchen waste is taken at 737 kg per m³ (Cornell Waste Management Institute, cafeteria waste). Grease-trap solids are [set], from the weekly cleaning, and kept a small share: worms are harmed above about 5 percent fat (Wu and colleagues, 2019).

3.2 Vessels, worm beds, and bays

Quantity	Calculation	Our example
A session, 20 days	$72\text{--}85 \text{ L} \times 20$	about 1.4–1.7 m ³
A year, 120 days	$72\text{--}85 \text{ L} \times 120$	about 8.7–10.2 m ³
Sealed, six weeks at peak	$85 \text{ L} \times 42 \text{ days}$	about 3.6 m ³

Quantity	Calculation	Our example
Vessels	$3.6 \text{ m}^3 \div 100 \text{ L}$	36 vessels of 100 L
Each worm bed, 4 months	a third of a year	about 2.9–3.4 m ³
Soil out, a year	$8.7\text{--}10.2 \text{ m}^3 \times 0.33\text{--}0.75$	about 2.9–7.6 m ³
Each bay, 6 months' soil	half a year out	about 1.4–3.8 m ³

The vessels. A vessel is opened no earlier than six weeks after it is sealed (paper, Section 10). The peak is sessions back to back: six weeks of the upper figure is about 3.6 m³, which is 36 vessels of 100 L. The paper's rule is to buy twice as many as the arithmetic suggests, so that nothing is opened early; 36 meets it when sessions do not run back to back, and is the minimum when they do. Fermented material keeps its volume (paper, Section 4).

The worm beds. Three, in rotation, as the paper has them; each takes about four months of the vessels' output, 2.9 to 3.4 m³ before the bulking and the finished vermicompost it is layered with. *Eisenia fetida* did not survive fresh faeces until they were layered with vermicompost (Yadav and colleagues, 2010), and fermented faeces without bulking were not accepted by worms (Factura and colleagues, 2010); the bulking share is [set] from the first batches, and sets each bed's size. Worms work best at 65 to 80 percent moisture (Lalander and colleagues, 2013).

The bays. Each bay takes six months of what the worm beds release and is then closed for a year: one filling, two maturing, three in all, as the paper has them. What leaves the worm beds is taken at `compost_volume_ratio`, 0.33 to 0.75, of what went in, until measured.

How long. Six weeks sealed, three to six months in the worms, up to six months filling a bay, and a full year closed: a year from the last addition, and up to about two years from the first. WHO gives 1.5 to 2 years at 2 to 20 °C for dry excreta with no new material added (Guidelines for the Safe Use of Wastewater, Excreta and Greywater, Vol. 4, 2006); the fermentation and the worms reduce bacteria but are not a barrier against *Ascaris* eggs (Harroff and colleagues, 2019; Andreev and colleagues, 2017), so the time and the trees carry the safety.

3.3 What the stream asks of the site, a year

Quantity	Calculation	Our example
Char at the seat	$11.3 \text{ L} \times 120$	about 1.4 m ³
Cover at the seat	$18.1 \text{ L} \times 120$	about 2.2 m ³
Carbohydrate, [assumed]	10 percent of the mass	about 870–1,020 kg

Quantity	Calculation	Our example
Soil to the trees	Section 3.2	about 2.9–7.6 m ³

The carbohydrate. With urine in the stream, its buffering holds the pH up. A laboratory study reached pH 4.1 to 5.2 only with at least 10 percent carbohydrate as molasses equivalent, and found char made little difference (Hübner and Moeller, Fermentation, 2026); a field trial on pit sludge used 10 percent molasses with 10 percent milk culture (Anderson and colleagues, 2015). The 10 percent of the mass sealed is [assumed], vessel contents taken at 1 kg a litre, and the kitchen's own starch counts toward it; the pH record replaces it.

4. Bill of materials

Quantities for the example. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
Clamp-lid vessels, 100 L	36	120 L; 60 L, lighter
Fermentation store, lockable	about 12 m ²	one timber store
Store lining, 50 mm foam	about 53 m ²	straw bales
Lactic starter	first season	whey after it
Molasses	870–1,020 kg a year	kitchen starch
Garden cart, two-wheel	1	none named
Wash stand with tank	1	a tap, if piped
Nitrile gloves	10 boxes of 100, a year	none named
Worm beds, covered	3	none named
Composting worms	4 boxes, [assumed]	none named
Maturation bays, covered	3, 1.4–3.8 m ³ each	none named
Sealed base	about 30 m ² , [assumed]	clay, rammed
Tarpaulins, 4 × 6 m	3	boards; old film

The siting, as designed. Every distance below is the larger of what the instruments in Section 10 ask, and every one is measured on the ground before anything is built; none has been measured yet.

From the store, beds, and bays to	At least
The spring, and never upslope of it	50 m
The stream, on a slope over 3°	50 m
The garden's pond, on a slope over 3°	50 m
Sleeping buildings and the dining hall	50 m
Other dwellings and summer kitchens	20 m
Any place where children are	20 m
The kitchen	20 m

The 50 m from sleeping buildings and the dining hall follows the children's-establishment regulation as reported for outdoor toilets; the enquiry settles whether it applies to a store.

The store, as designed. Shaded, ventilated, lockable, lined against frost, with the vessels on a level floor and room to open one without moving the others. Each vessel labelled with the date it was sealed. A wash stand at the door. The route from the toilets short, level, and apart from where the day's work and play go.

5. Build

Step	Trade	Time	If done out of order
Written enquiry answered	household	[from L2]	nothing is bought
Walk; distances measured	household	[from L2]	store in the strip
Sealed base poured	builder	[from L2]	leachate soaks away
Store set up and lined	builder	[from L2]	vessels freeze
Worm beds and bays built	carpenter	[from L2]	[from L2]
Worms stocked, bedded	responsible adult	[from L2]	worms die
Vessels, starter, cover	responsible adult	[from L2]	[from L2]

Step	Trade	Time	If done out of order
Written procedure	household	[from L2]	no answer at inspection

The worm beds are stocked and running on garden material before the first fermented vessel is opened, so that the worms meet fermented faeces only layered into a living bed.

6. Cost

Estimates, not quotations: each line is either a price listed by a Ukrainian retailer, seen on 2026-09-29, or the sanitation paper's own estimate for autumn 2026, marked "paper, at 25". Place: not a quotation; western Ukrainian listings where found, national otherwise. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

6.1 Fermentation

Item	UAH	EUR	Basis
Vessels, 36 × 100 L at 1,070	38,520	741	estimated, listing
Store, two steel sheds 2 × 3 m	72,000	1,385	estimated, listing
Lining, 106 foam sheets at 56	5,936	114	estimated, listing
Lactic starter, first season	6,000	115	paper, at 25
Subtotal	122,456	2,355	

6.2 Handling

Item	UAH	EUR	Basis
Garden cart, two-wheel	2,960	57	estimated, listing
Wash stand, 10 L tank	3,371	65	estimated, listing
Nitrile gloves, 10 boxes at 216	2,160	42	estimated, listing
Subtotal	8,491	163	

6.3 Worm beds and bays

Item	UAH	EUR	Basis
Worm beds, three, covered	45,000	865	paper, at 25
Maturation bays, three	30,000	577	paper, at 25
Sealed base, 3 m³ of M200	10,950	211	listing; area assumed
Composting worms, 4 boxes	1,600	31	listing; qty assumed
Tarpaulins, 3 at 913.50	2,741	53	estimated, listing
Subtotal	90,291	1,736	

6.4 Summary

Section	UAH	EUR
Fermentation	122,456	2,355
Handling	8,491	163
Worm beds and bays	90,291	1,736
Construction subtotal	221,238	4,255
Design and sanitary review, 8 percent	17,700	340
Contingency, 20 percent	44,200	850
Total external cost	283,138	5,445

Against the paper's 136,000 UAH for the same stages at 25 people, the difference is the store, now sized for 36 vessels and lined against frost, and the sealed base. Three lines are still the paper's estimates at 25 — the starter, the worm beds, and the bays — and are marked for a new estimate from the sizes in Section 3.2; the total does not yet include the difference. The percentages are the paper's own.

Contribution, not in the total. None counted. The responsible adult's work is a cost, not a contribution.

Running cost. Molasses, about 870 to 1,020 kg a year at a listed 250 UAH for 7 kg: 30,912 to 36,338 UAH a year, [assumed] rate, until the pH record sets it — the largest running line. Cover, about 2.2 m³ a year: from the site, or sawdust at a listed 400 UAH per m³, about 872 UAH. Whey from the household's

dairy after the first season, or at a listed 20.50 UAH a litre, dose [set]. Gloves, 2,160 UAH a year. Char from the biochar module. The responsible adult's hours, [set], paid at no less than the legal minimum wage, 8,647 UAH a month in 2026, for the hours worked.

7. Operation and maintenance

Responsible: one named adult, paid, with a deputy — [from L2]. Not a rota.

- **Daily, in session:** buckets exchanged by the responsible adult and taken to the store on the cart; tipped into the open vessel; the vessel topped with char and cover and closed; buckets rinsed at the dedicated point, the rinse to the septic tank; the kitchen's cooked food and plate waste into the same vessel, covered.
- **As each vessel fills:** starter and carbohydrate added, pH read, the vessel sealed and labelled with the date; not opened for six weeks.
- **Weekly:** the store checked; the grease-trap solids added in small measure; the char and cover stocks checked.
- **Monthly:** worm beds fed in thin layers of fermented material with bulking and finished vermicompost; moisture felt and kept between 65 and 80 percent; beds kept in strict rotation.
- **Seasonal:** worm beds covered and banked before frost; a bay closed and labelled with its closing date when it has taken six months.
- **Annual:** a bay opened only a full year after its closing date; the soil taken to trees, hedgerows, windbreaks, and nursery stock, never to a vegetable bed, and not in the month before a fruit harvest; the batch record kept — sealed, opened, bedded, closed, applied.

The making of the soil is visible and explicable; the handling stays with the responsible adult. The worm beds may be shown; open vessels are not, and the store stays locked. How the programme relates to this work is described in the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Vessel opened early	label date	reseal; restart the six weeks
pH stays high	pH record	more carbohydrate; [from L2]
Worms die or leave	[from L2]	more bulking; less fat

Failure	Early sign	Repair and cost
Worm bed floods	[from L2]	more cover; drain the bed
Bay opened early	label date	reclose; restart the year
Leachate toward water	[from L2]	base, covers, distance
Worm beds freeze	[from L2]	banking; [from L2]
Receiving ground too small	[from L2]	more trees; fewer vessels
Inspection refuses	written answer	the plumbed-units case

The paper names the questions an inspector will ask — smell, flies, hand hygiene, and what happens to the material — and answers them with sealed vessels, a working vent, handwashing at the cubicle door, a written procedure, and the rule of the year.

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-09-29, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Instruments found on 2026-09-29 and read on secondary copies and official summaries, because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked.

- **MOH Order No. 145 of 17.03.2011**, the state sanitary rules for maintaining settlement territories. Point 2.21: liquid waste — "фекалії, сеча, помий", faeces, urine, slops — may be kept in cesspits, and placing it on private land or using it as fertiliser is prohibited. The point is written for cesspit contents and says nothing of dry toilets or composting, but is the rule an inspector would most likely apply by analogy, and a stream that keeps its urine reads closer to its words. Point 2.22: cesspits at least 20 m from the plot boundaries of children's and medical establishments, dwellings, public buildings, and play areas, and at least 50 m from public wells and captured springs.
- **MOH Order No. 377 of 20.03.2026**, the sanitary regulation for children's health and recreation establishments, read in summaries only: toilets at least 50 m from sleeping buildings and the dining hall; waste containers at least 20 m from places where children are. Whether the programme is such an establishment is for the enquiry.
- **Law No. 2320-IX of 20.06.2022**, on waste management: no exclusion for faeces, unlike the EU directive; bucket contents that are not sewage are arguably waste under it.
- **ДСТУ 7369:2013** covers sewage sludge from treatment plants, not dry-toilet material.
- **Water Code, No. 213/95-BP of 06.06.1995**, Arts. 88–89, and **ДСанПіН 2.2.4-171-10**, point 3.32: as in the compost module — a protective strip of 25 m, 50 m on a slope over 3°, along the stream and the pond, with no store of waste in it; a spring capture at least 50 m from latrines and other places that pollute soil.

The written enquiry to the district office of the state food-safety and consumer-protection service, with the design drawn and the procedure written, asks: whether point 2.21 applies to this stream; whether matured material may go on the site's own trees; whether Order No. 377 applies to the programme; and which distances bind. Nothing is bought until it is answered.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A drier valley:** less cover is needed to hold the stream; the worm beds need watering, from rainwater, never the spring.
- **A colder valley:** the store and the worm beds are the weak points; more lining, beds against a heated wall, and a longer year.
- **A smaller group:** fewer vessels, the same three beds and bays, smaller; the store shrinks with the vessels.

- **A larger group:** vessels, beds, and bays scale with Section 3; the receiving ground must grow with them.
- **Plumbed units required:** the bucket figures fall with bucket_share; the kitchen's share stays.
- **Urine diverted, where a site chooses it:** the vessels take a third of the volume, and the cover and the carbohydrate fall with it.

12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]
- **From whose knowledge:** the sanitation working paper, EN v0.2; Factura, Bettendorf, Buzie, Pieplow, Reckin, and Otterpohl, Terra Preta Sanitation, Water Science and Technology, 2010, for the sequence of fermentation and worms; Joseph Jenkins, Humanure Sanitation, 2009, for the carbon cover; Rose and colleagues, 2015, for excreta; WHO, 2006, for maturation and crop restriction; Yadav and colleagues, 2010, Lalander and colleagues, 2013, and Wu and colleagues, 2019, for the worm beds; Anderson and colleagues, 2015, and Hübner and Moeller, 2026, for the carbohydrate; Harroff and colleagues, 2019, and Andreev and colleagues, 2017, for what fermentation does not do; Bischak and colleagues, 2024, and Fidel and colleagues, 2018, for char and nitrogen; Silvennoinen and colleagues, 2019, for kitchen waste; Ukrainian retail listings of 2026-09-29 for prices.
- **Traditional knowledge:** none. What the valley did with a household's waste is for the elders; the literature found nothing.
- **Text:** CC BY-SA 4.0
- **Drawings and designs:** [hardware licence — open decision]
- **Measurement data:** none yet
- **Gate 1:** household decision, 2026-09-29, recorded by Michel Garand — v0.2 may be published as it stands, in English, Ukrainian, and German
- **War-time review:** 2026-09-29, Michel Garand — v0.2 read against compendium Section 2; nothing places the site; the note on permits and local rules added; accepted

13. What to check

- **The written answer** of the district sanitary service, before anything is bought: point 2.21, Order No. 377, the distances.
- **The distances** — every row of the siting table, measured with a tape on the walk; the slope beside the stream and the pond.
- **bucket_share** — the uses the buckets carry, from the first sessions; and the plumbed-units case, if it comes.

- **cover_at_seat** — the cover a use takes, from the first week.
- **meals_per_day** and the kitchen's waste — weighed for a session; the grease-trap solids from the weekly cleaning.
- **The carbohydrate** — the pH at sealing and opening, replacing the assumed 10 percent.
- **The worm beds' bulking share** — from the first batches; it sets each bed's size.
- **compost_volume_ratio** for the worm beds — measured in and out.
- **The receiving ground** — trees, hedgerows, windbreaks, and nursery beds counted on the walk against 2.9 to 7.6 m³ a year.
- **Three lines at 25** — the starter, the worm beds, and the bays, re-estimated from Section 3.2.
- **The sealed base** — its area from the layout, replacing the assumed 30 m².
- **The responsible adult** — hours and pay.
- **The Ukrainian row** — every instrument read and signed.
- **Quotations** for the lines in Section 6, dated, to replace the listings and estimates.
- **From L2, once built:** build times and out-of-order consequences in Section 5; the daily record in Section 7; early signs and repair costs in Section 8; Section 9.

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